

Khwaja Yunus Ali University

School of Science and Engineering

Department of Computer Science and Engineering

Mid-term Examination: Fall Semester 2020

Program: B.Sc. in CSE (Batch – 10th) 1st Year 2nd Semester

Course Title: Physics-II

Course Code: PHY 1201

Time: 10 Minutes

Total Marks: 10 X 1 = 10

Student ID:	Batch:	Session:
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Part 1: MCQ

These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.

- Another name of the Brownian motion is
a) zigzag b) elastic motion c) pedesis
d) rotatory e) linear motion
- Which is the unit of latent heat?
a) JK^{-1} b) Jkg^{-1} c) $Jkg^{-1}K^{-1}$
d) J e) K
- For Adiabatic process,
a) $T=\text{Constant}$ b) $V=\text{Constant}$ c) $P=\text{Constant}$
d) $S=\text{Constant}$ e) $G=\text{Constant}$
- What is the value of the Boltzmann constant?
a) $1.2 \times 10^{-15} JK^{-1}$ b) $1.31 \times 10^{-15} JK^{-1}$ c) $1.31 \times 10^{23} JK^{-1}$
d) $1.38 \times 10^{23} JK^{-1}$ e) $1.38 \times 10^{-23} JK^{-1}$
- The difference between two molar specific heats equal to
a) molar gas constant b) heat c) Boltzmann constant
d) Plank constant e) γ
- Angular momentum of an electron in the 1st orbit of atom is equal to
a) $h/2\pi$ b) h/π c) $2h/\pi$
d) $nh/2\pi$ e) nh/π
- Centrifugal force of an electron in rotation around the nucleus comes from
a) Newtonian force b) gravitation c) acceleration
d) central force e) Colombian force
- What is the radius of Hydrogen nucleus?
a) 3.7fm b) 2.7fm c) 1.7fm
d) 1.2fm e) 0.53fm
- Which one is the alpha particle?
a) 1_1H b) 2_1H c) 3_1H
d) 1_0n e) 4_2He
- How many energy releases in every fission reaction?
a) 13.6MeV b) 37MeV c) 200MeV
d) 931.5MeV e) 1330MeV

P.T.O.

Part 2: SAQ

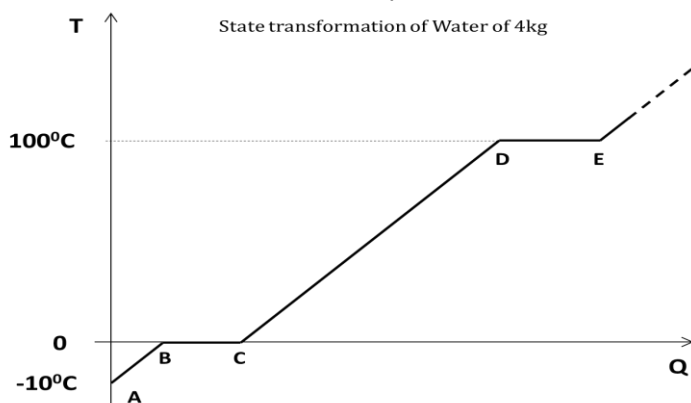
Time: 1 Hour and 20 Minutes

Total Marks: 20

(Answer Question No. 1 and any three from the rest.)

[The figures in the right margin indicate full marks for the respective question]

1. See the transformation curve and answer the questions that follows:



- a) What is entropy? 1
 - b) Find the value of BC. 1
 - c) Compare the change in entropy for CD and DE portion of the above-mentioned curve. 3

- a) Differentiate real and Ideal gases graphically and justify your answer. 2
 - b) Construct the Van-der-wall equation in the correction of real gas equation. 3

- a) Write a short note on equipartition of energy. 2
 - b) The mean free path of the molecules of a gas is $6 \times 10^{-8}\text{m}$ and the diameter of a molecule is $2.5 \times 10^{-10}\text{m}$. Find the number of molecules per cm^3 . 3

- a) Explain the principle of the unattainability of absolute zero. 2
 - b) Which of these two is the best engine? Present your argument in favor of the answer. 3

Engine	Source Temperature ($^{\circ}\text{C}$)	Sink Temperature ($^{\circ}\text{C}$)	Absorbed Heat (J)	Rejected Heat (J)
Engine-I	327	27	300	600
Engine-II	527	127	1800	900

- a) What are the main components of nuclear reactor? 2
 - b) Illustrate how a nuclear reaction in a nuclear reactor expands. 3

- a) Define 1 a.m.u. 1
 - b) Draw the spectral series of hydrogen atom. 2
 - c) Calculate the energy of an electron of 2nd orbit of Helium atom. 2